

Jeff Shen

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Education

H.B.Sc., Statistics, Astronomy & Astrophysics, Mathematics
University of Toronto

2018-2022

Technical Skills

Data: Python, NumPy, SciPy, Pandas, PyTorch, Scikit-Learn, Matplotlib, Stan, Seaborn, SQL, NoSQL, R
Web: HTML5, CSS3, JavaScript, React (Native), Redux, Node, Express.js, Flask

Other: Rust, Docker, shell scripting, vim, version control (git), cloud computing (AWS, Azure, Google Cloud Platform), Unix/Linux, SymPy, Astropy, Selenium, NLTK, $\text{\LaTeX}$, regular expressions

Research Experience

- 2020 Aug. - Present **Undergraduate Research Course**
Canadian Institute for Theoretical Astrophysics
Supervised by [Prof. Gwendolyn Eadie](#) and [Prof. Norman Murray](#). Study of mass distribution of the Milky Way in a Bayesian framework.
- 2020 May - 2020 Aug. **Summer Undergraduate Research Program**
Dunlap Institute for Astronomy and Astrophysics, University of Toronto
Supervised by [Dr. Allison Man](#). Study of molecular gas in a strongly-lensed protocluster system at $z \approx 2.9$. Awarded best poster in student competition.

Honours and Awards

- 2020 **Summer Undergraduate Research Program Award (\$9,500 CAD)**
Dunlap Institute for Astronomy and Astrophysics, University of Toronto

Publications

- 2020 October **Jeff Shen**, Allison W. S. Man, Johannes Zabl, Zhi-Yu Zhang, Mikkel Stockmann, Gabriel Brammer, and Katherine E. Whitaker. "Molecular gas in a gravitationally lensed galaxy group at $z = 2.9$ ", submitted to *ApJ*.

Independent Coursework

- 2020 **Practical Time Series Analysis**, The State University of New York (Coursera)
Stationarity, differencing, ACF, PACF, autocovariance. MA processes, AR processes, AR-MA duality. Backshift operator, difference equations, Yule-Walker matrix equations. AIC, SSE, model selection. ARMA and ARIMA models. Seasonality, SARIMA. Exponential smoothing, forecasting.
- 2019 **Data-Driven Astronomy**, The University of Sydney (Coursera)
Pulsars, black holes, exoplanets, stellar evolution, distances, redshift, galaxy classification. Image stacking, algorithm analysis, crossmatching, databases, regression, classification, machine learning. Implementation in Python with Astropy; SQL.
- 2019 **Intro to Deep Learning with PyTorch**, Udacity
Mathematics of neural networks. Data preparation, regularization techniques. Image classification with CNNs and OpenCV. Time series with RNNs. Implementation in Python with PyTorch.
- 2018 **Intro to Machine Learning**, Udacity
Classifiers (naïve Bayes, SVMs, ensemble methods), data processing, feature selection, regression, clustering, outlier detection, PCA, evaluation metrics. Implementation in Python with scikit-learn.

Projects

- 2020 **Parallelized ray tracer**
Implemented a parallelized ray tracer from mathematical first principles in Rust.
- 2020 **Markov Chain Monte Carlo in Rust**
Implemented a Markov chain Monte Carlo sampler from scratch in Rust. Built on a custom statistics crate.
- 2019 **Image classification with CNNs**
Cleaned a dataset of several thousands of images of flowers. Optimized a deep learning model with PyTorch and applied cutting-edge techniques to achieve 99% accuracy in the classification of species of flowers. Could potentially be adapted for classification of astronomical objects.
- 2018 **Analysis of global Internet usage statistics**
Spearheaded the programming aspect of a collaborative statistical analysis project. Proposed research questions and used R to develop data-backed answers. Gave a presentation of results.

Outreach

- 2019 - Present **Space Science Journalist (Current in Space segment)**, [*The Star Spot* podcast](#)
Curate, write and record stories about the latest research and news in astronomy.
Episodes 177 - present.

Volunteering

- 2015 - 2018 **Python Workshop Lead**, Sir Winston Churchill Secondary, Vancouver, B.C.
Coordinated and delivered workshops to prepare students for national programming contest.

Interests

Languages: English (native), French, Chinese, Norwegian (basic)

Other: *nix customization, computer graphics, 3D modelling, running, baking, reading, guitar